DHAKA UNIVERSITY OF ENGINEERING & TECHNOLOGY

DUET ADMISSION PREPARATION

C & C++ Programs

মৌলিক প্রোগ্রামসমূহ — পাশাপাশি তুলনা

C LANGUAGE   C++ LANGUAGE

60 Programs · Side by Side · Bangla Explanation

STUDENT INFORMATION

STUDENT NAME	STUDENT ID	BATCH

DEPARTMENT	POLYTECHNIC INSTITUTE	ACADEMIC YEAR

গাণিতিক অপারেশন — Arithmetic Operations

০১. ক্যালকুলেটর — যোগ, বিয়োগ, গুণ ও ভাগ 01. SUM, SUBTRACTION, MULTIPLICATION & DIVISION OF TWO NUMBERS

নমুনা আউটপুট:
```
Enter two numbers: 10 3
Sum = 13.000000,  Subtraction = 7.000000
Multiplication = 30.000000,  Division = 3.333333
b=0 হলে → "Division not possible" প্রিন্ট হবে।
```

C
```c
#include <stdio.h>

int main() {
    float a, b;
    printf("Enter two numbers: ");
    scanf("%f %f", &a, &b);
    printf("Sum = %f\n", a + b);
    printf("Subtraction = %f\n", a - b);
    printf("Multiplication = %f\n", a * b);
    if (b != 0)
        printf("Division = %f\n", a / b);
    else
        printf("Division not possible (b=0)\n");
    return 0;
}
```

C++
```cpp
#include <iostream>
using namespace std;

int main() {
    float a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Sum = " << a + b << endl;
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    if (b != 0)
        cout << "Division = " << a / b << endl;
    else
        cout << "Division not possible (b=0)" << endl;
    return 0;
}
```

তুলনামূলক অপারেশন — Comparisons

০২. দুটি সংখ্যার মধ্যে বড়টি খোঁজা 02. FIND LARGER OF TWO NUMBERS

```
if (a > b) → a বড়।  else if (b > a) → b বড়।  else → দুটো সমান।
  উদাহরণ: a=8, b=3 → "Larger number is: 8"
          a=5, b=5 → "Both numbers are equal."
```

C
```c
#include <stdio.h>

int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    if (a > b)
        printf("Larger number is: %d\n", a);
    else if (b > a)
        printf("Larger number is: %d\n", b);
    else
        printf("Both numbers are equal.\n");
    return 0;
}
```

C++
```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    if (a > b)
        cout << "Larger number is: " << a << endl;
    else if (b > a)
        cout << "Larger number is: " << b << endl;
    else
        cout << "Both numbers are equal." << endl;
    return 0;
}
```

০৩. দুটি সংখ্যার মধ্যে ছোটটি খোঁজা 03. FIND SMALLER OF TWO NUMBERS

```
if (a < b) → a ছোট।  else if (b < a) → b ছোট।  else → দুটো সমান।
  উদাহরণ: a=3, b=8 → "Smaller number is: 3"
          a=5, b=5 → "Both numbers are equal."
```

C
```c
#include <stdio.h>

int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    if (a < b)
        printf("Smaller number is: %d\n", a);
    else if (b < a)
        printf("Smaller number is: %d\n", b);
    else
        printf("Both numbers are equal.\n");
    return 0;
}
```

C++
```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    if (a < b)
        cout << "Smaller number is: " << a << endl;
    else if (b < a)
        cout << "Smaller number is: " << b << endl;
    else
        cout << "Both numbers are equal." << endl;
    return 0;
}
```

০৪. তিনটি সংখ্যার মধ্যে সবচেয়ে বড় (টার্নারি অপারেটর দিয়ে) 04. FIND LARGEST OF THREE NUMBERS (CONDITIONAL OPERATOR)

টার্নারি: (শর্ত) ? সত্য-মান : মিথ্যা-মান
max = (a>b && a>c) ? a : (b>c ? b : c)
উদাহরণ: a=4, b=9, c=6 → max = 9

C

```c
#include <stdio.h>

int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    int max = (a > b && a > c) ? a : (b > c ? b : c);
    printf("Largest number is: %d\n", max);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    int max = (a > b && a > c) ? a : (b > c ? b : c);
    cout << "Largest number is: " << max << endl;
    return 0;
}
```

০৫. তিনটি সংখ্যার মধ্যে সবচেয়ে ছোট (টার্নারি অপারেটর দিয়ে) 05. FIND SMALLEST OF THREE NUMBERS (CONDITIONAL OPERATOR)

টার্নারি: (শর্ত) ? সত্য-মান : মিথ্যা-মান
min = (a<b && a<c) ? a : (b<c ? b : c)
উদাহরণ: a=4, b=9, c=6 → min = 4

C

```c
#include <stdio.h>

int main() {
    int a, b, c;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);
    int min = (a < b && a < c) ? a : (b < c ? b : c);
    printf("Smallest number is: %d\n", min);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    int min = (a < b && a < c) ? a : (b < c ? b : c);
    cout << "Smallest number is: " << min << endl;
    return 0;
}
```

০৬. তিনটি সংখ্যার গড় 06. AVERAGE OF THREE NUMBERS

গড় = (a + b + c) / 3   → float ব্যবহার করা হয় দশমিক পেতে।
উদাহরণ: a=4, b=8, c=9 → গড় = (4+8+9)/3 = 21/3 = 7.000000

C

```c
#include <stdio.h>

int main() {
    float a, b, c;
    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);
    float avg = (a + b + c) / 3;
    printf("Average = %f\n", avg);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    float a, b, c;
    cout << "Enter three numbers: ";
    cin >> a >> b >> c;
    float avg = (a + b + c) / 3;
    cout << "Average = " << avg << endl;
    return 0;
}
```

জ্যামিতি ও ক্ষেত্রফল — Geometry & Area

০৭. সমকোণী ত্রিভুজের ক্ষেত্রফল 07. AREA OF A RIGHT-ANGLED TRIANGLE

C

```c
#include <stdio.h>

int main() {
    float base, height;
    printf("Enter base and height: ");
    scanf("%f %f", &base, &height);
    float area = 0.5 * base * height;
    printf("Area of right-angled"
           " triangle = %f\n", area);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    float base, height;
    cout << "Enter base and height: ";
    cin >> base >> height;
    float area = 0.5 * base * height;
    cout << "Area of right-angled"
         << " triangle = " << area << endl;
    return 0;
}
```

০৮. বিষমবাহু ত্রিভুজের ক্ষেত্রফল (হেরোনের সূত্র) 08. AREA OF A SCALENE TRIANGLE (HERON'S FORMULA)

হেরোনের সূত্র: তিন বাহু a, b, c জানা থাকলে ক্ষেত্রফল বের করা যায়।
আধা-পরিসীমা: s = (a + b + c) / 2
ক্ষেত্রফল = √(s × (s-a) × (s-b) × (s-c))
উদাহরণ: a=3, b=4, c=5 → s=6 → area = √(6×3×2×1) = 6.00

C

```c
#include <stdio.h>
#include <math.h>

int main() {
    float a, b, c;
    printf("Enter three sides: ");
    scanf("%f %f %f", &a, &b, &c);
    float s = (a + b + c) / 2;
    float area = sqrt(s * (s - a) * (s - b) * (s - c));
    printf("Area of scalene"
           " triangle = %f\n", area);
    return 0;
}
```

C++

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    float a, b, c;
    cout << "Enter three sides: ";
    cin >> a >> b >> c;
    float s = (a + b + c) / 2;
    float area = sqrt(s * (s - a) * (s - b) * (s - c));
    cout << "Area of scalene"
         << " triangle = " << area << endl;
    return 0;
}
```

০৯. বৃত্তের ক্ষেত্রফল 09. AREA OF A CIRCLE

বৃত্তের ক্ষেত্রফল সূত্র: A = π × r²
#define PI 3.1416 দিয়ে π ধ্রুবক সংজ্ঞায়িত করা হয়।
উদাহরণ: r=7 → A = 3.1416 × 7 × 7 = 153.94

C

```c
#include <stdio.h>
#define PI 3.1416

int main() {
    float radius;
    printf("Enter radius: ");
    scanf("%f", &radius);
    float area = PI * radius * radius;
    printf("Area of circle = %f\n", area);
    return 0;
}
```

C++

```cpp
#include <iostream>
#define PI 3.1416
using namespace std;

int main() {
    float radius;
    cout << "Enter radius: ";
    cin >> radius;
    float area = PI * radius * radius;
    cout << "Area of circle = " << area << endl;
    return 0;
}
```

সংখ্যার বৈশিষ্ট্য — Number Properties

১০. ধনাত্মক, ঋণাত্মক নাকি শূন্য 10. CHECK POSITIVE, NEGATIVE, OR ZERO

n > 0 → Positive | n < 0 → Negative | n = 0 → Zero
উদাহরণ: 5 → Positive | -3 → Negative | 0 → Zero

C

```c
#include <stdio.h>

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    if (n > 0)
        printf("Positive\n");
    else if (n < 0)
        printf("Negative\n");
    else
        printf("Zero\n");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n > 0)
        cout << "Positive" << endl;
    else if (n < 0)
        cout << "Negative" << endl;
    else
        cout << "Zero" << endl;
    return 0;
}
```

১১. বিজোড় না জোড় 11. CHECK ODD OR EVEN

num % 2 == 0 → জোড় | num % 2 ≠ 0 → বিজোড়
উদাহরণ: 4 → Even (4%2=0) | 7 → Odd (7%2=1)

C

```c
#include <stdio.h>

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    if (num % 2 != 0)
        printf("The number is odd.");
    else
        printf("The number is even.");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;
    if (num % 2 != 0)
        cout << "The number is odd.";
    else
        cout << "The number is even.";
    return 0;
}
```

১২. অধিবর্ষ (লিপ ইয়ার) যাচাই 12. CHECK LEAP YEAR

লিপ ইয়ার শর্ত:
✓ 4 দিয়ে বিভাজ্য এবং 100 দিয়ে বিভাজ্য নয় → লিপ ইয়ার
✓ 400 দিয়ে বিভাজ্য → লিপ ইয়ার
উদাহরণ: 2024 ✓ লিপ | 1900 ✗ লিপ নয় | 2000 ✓ লিপ

C

```c
#include <stdio.h>

int main() {
    int year;
    printf("Enter a year: ");
    scanf("%d", &year);
    if ((year % 4 == 0 && year % 100 != 0)
        || (year % 400 == 0))
        printf("Leap year\n");
    else
        printf("Not a leap year\n");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int year;
    cout << "Enter a year: ";
    cin >> year;
    if ((year % 4 == 0 && year % 100 != 0)
        || (year % 400 == 0))
        cout << "Leap year" << endl;
    else
        cout << "Not a leap year" << endl;
    return 0;
}
```

তাপমাত্রা রূপান্তর — Temperature Conversions

১৩. সেলসিয়াস থেকে ফারেনহাইট 13. CELSIUS TO FAHRENHEIT

সেলসিয়াস → ফারেনহাইট: F = (C × 9/5) + 32
উদাহরণ: C=0° → F=32° (বরফ গলনাঙ্ক)
C=100° → F=212° (পানি স্ফুটনাঙ্ক)
C=37° → F=98.6° (মানবদেহের তাপমাত্রা)

C

```c
#include <stdio.h>

int main() {
    float celsius, fahrenheit;
    printf("Enter temperature in Celsius: ");
    scanf("%f", &celsius);
    fahrenheit = (celsius * 9 / 5) + 32;
    printf("Temp in Fahrenheit: %f\n",
           fahrenheit);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    float celsius, fahrenheit;
    cout << "Enter temperature in Celsius: ";
    cin >> celsius;
    fahrenheit = (celsius * 9 / 5) + 32;
    cout << "Temp in Fahrenheit: "
         << fahrenheit << endl;
    return 0;
}
```

১৪. ফারেনহাইট থেকে সেলসিয়াস 14. FAHRENHEIT TO CELSIUS

```
ফারেনহাইট → সেলসিয়াস:  C = (F - 32) × 5/9
  উদাহরণ: F=32°   → C=0°   (বরফ গলনাঙ্ক)
          F=212°  → C=100° (পানি স্ফুটনাঙ্ক)
          F=98.6° → C=37°  (মানবদেহের তাপমাত্রা)
```

C

```c
#include <stdio.h>

int main() {
    float fahrenheit, celsius;
    printf("Enter temperature in Fahrenheit: ");
    scanf("%f", &fahrenheit);
    celsius = (fahrenheit - 32) * 5 / 9;
    printf("Temperature in Celsius: %f\n", celsius);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    float fahrenheit, celsius;
    cout << "Enter temperature in Fahrenheit: ";
    cin >> fahrenheit;
    celsius = (fahrenheit - 32) * 5 / 9;
    cout << "Temp in Celsius: "
         << celsius << endl;
    return 0;
}
```

অক্ষর অপারেশন — Character Operations

১৫. বড় হাতের না ছোট হাতের অক্ষর 15. CHECK UPPERCASE OR LOWERCASE

```
'a'-'z' → Lowercase  |  'A'-'Z' → Uppercase  |  অন্যটা → Not an alphabet
  উদাহরণ: 'g' → Lowercase  |  'G' → Uppercase  |  '3' → Not an alphabet
```

C

```c
#include <stdio.h>

int main() {
    char ch;
    printf("Enter a character: ");
    scanf("%c", &ch);
    if (ch >= 'a' && ch <= 'z')
        printf("Lowercase letter\n");
    else if (ch >= 'A' && ch <= 'Z')
        printf("Uppercase letter\n");
    else
        printf("Not an alphabet letter\n");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;
    if (ch >= 'a' && ch <= 'z')
        cout << "Lowercase letter" << endl;
    else if (ch >= 'A' && ch <= 'Z')
        cout << "Uppercase letter" << endl;
    else
        cout << "Not an alphabet letter" << endl;
    return 0;
}
```

১৬. স্বরবর্ণ না ব্যঞ্জনবর্ণ — if-else দিয়ে 16. CHECK VOWEL OR CONSONANT (IF-ELSE)

```
tolower() দিয়ে ছোট হাতে করা হয়, তারপর a/e/i/o/u → Vowel, বাকি → Consonant।
  উদাহরণ: 'A' → tolower → 'a' → Vowel  |  'B' → Consonant  |  '5' → Not an alphabet
```

C

```c
#include <stdio.h>
#include <ctype.h>

int main() {
    char ch;
    printf("Enter an alphabet: ");
    scanf("%c", &ch);
    ch = tolower(ch);
    if (ch == 'a' || ch == 'e' || ch == 'i'
        || ch == 'o' || ch == 'u')
        printf("Vowel\n");
    else if (ch >= 'a' && ch <= 'z')
        printf("Consonant\n");
    else
        printf("Not an alphabet\n");
    return 0;
}
```

C++

```cpp
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char ch;
    cout << "Enter an alphabet: ";
    cin >> ch;
    ch = tolower(ch);
    if (ch == 'a' || ch == 'e' || ch == 'i'
        || ch == 'o' || ch == 'u')
        cout << "Vowel" << endl;
    else if (ch >= 'a' && ch <= 'z')
        cout << "Consonant" << endl;
    else
        cout << "Not an alphabet" << endl;
    return 0;
}
```

১৭. স্বরবর্ণ না ব্যঞ্জনবর্ণ — switch-case দিয়ে 17. CHECK VOWEL OR CONSONANT (SWITCH-CASE)

switch-case এ a,e,i,o,u → Vowel। default এ 'a'-'z' হলে Consonant, না হলে Not an alphabet। P16 এর মতোই কাজ — শুধু if-else এর বদলে switch-case ব্যবহার করা হয়েছে।

C

```c
#include <stdio.h>
#include <ctype.h>

int main() {
    char ch;
    printf("Enter an alphabet: ");
    scanf("%c", &ch);
    ch = tolower(ch);
    switch(ch) {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
            printf("Vowel\n");
            break;
        default:
            if (ch >= 'a' && ch <= 'z')
                printf("Consonant\n");
            else
                printf("Not an alphabet\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char ch;
    cout << "Enter an alphabet: ";
    cin >> ch;
    ch = tolower(ch);
    switch(ch) {
        case 'a':
        case 'e':
        case 'i':
        case 'o':
        case 'u':
            cout << "Vowel" << endl;
            break;
        default:
            if (ch >= 'a' && ch <= 'z')
                cout << "Consonant" << endl;
            else
                cout << "Not an alphabet" << endl;
    }
    return 0;
}
```

লুপ — Loops

১৮. "Bangladesh" ১০ বার প্রিন্ট (for / while / do-while) 18. PRINT "BANGLADESH" 10 TIMES (FOR / WHILE / DO-WHILE)

C

```c
#include <stdio.h>

int main() {
    // Using for loop
    printf("Using for loop:\n");
    for (int i = 0; i < 10; i++)
        printf("Bangladesh\n");

    // Using while loop
    printf("\nUsing while loop:\n");
    int j = 0;
    while (j < 10) {
        printf("Bangladesh\n");
        j++;
    }

    // Using do-while loop
    printf("\nUsing do-while loop:\n");
    int k = 0;
    do {
        printf("Bangladesh\n");
        k++;
    } while (k < 10);

    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    // Using for loop
    cout << "Using for loop:" << endl;
    for (int i = 0; i < 10; i++)
        cout << "Bangladesh" << endl;

    // Using while loop
    cout << "\nUsing while loop:" << endl;
    int j = 0;
    while (j < 10) {
        cout << "Bangladesh" << endl;
        j++;
    }

    // Using do-while loop
    cout << "\nUsing do-while loop:" << endl;
    int k = 0;
    do {
        cout << "Bangladesh" << endl;
        k++;
    } while (k < 10);

    return 0;
}
```

১৯. ১ থেকে ১০ পর্যন্ত প্রিন্ট (for / while / do-while) 19. PRINT 1 TO 10 (FOR / WHILE / DO-WHILE)

C

```c
#include <stdio.h>

int main() {
    // Using for loop
    printf("Using for loop:\n");
    for (int i = 1; i <= 10; i++)
        printf("%d ", i);

    // Using while loop
    printf("\nUsing while loop:\n");
    int j = 1;
    while (j <= 10) {
        printf("%d ", j);
        j++;
    }

    // Using do-while loop
    printf("\nUsing do-while loop:\n");
    int k = 1;
    do {
        printf("%d ", k);
        k++;
    } while (k <= 10);

    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    // Using for loop
    cout << "Using for loop:" << endl;
    for (int i = 1; i <= 10; i++)
        cout << i << " ";

    // Using while loop
    cout << "\nUsing while loop:" << endl;
    int j = 1;
    while (j <= 10) {
        cout << j << " ";
        j++;
    }

    // Using do-while loop
    cout << "\nUsing do-while loop:" << endl;
    int k = 1;
    do {
        cout << k << " ";
        k++;
    } while (k <= 10);

    return 0;
}
```

২০. ১ থেকে ১০ পর্যন্ত যোগফল (for লুপ) 20. SUM OF 1 TO 10 (FOR LOOP)

C

```c
#include <stdio.h>

int main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++)
        sum = sum + i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++)
        sum = sum + i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

২১. মৌলিক সংখ্যা যাচাই 21. CHECK PRIME NUMBER

মৌলিক সংখ্যা: শুধু ১ ও নিজে দিয়ে বিভাজ্য।
পদ্ধতি: ২ থেকে (num-1) পর্যন্ত ভাগ করে দেখা হয়।
উদাহরণ: 7 → 2,3,4,5,6 দিয়ে ভাগ হয় না → মৌলিক ✓
9 → 3 দিয়ে ভাগ হয় → মৌলিক নয় ✗

C

```c
#include <stdio.h>

int main() {
    int num, count = 0;
    printf("Enter any positive number: ");
    scanf("%d", &num);
    if (num <= 1) {
        printf("Not Prime Number");
        return 0;
    }
    for (int i = 2; i < num; i++) {
        if (num % i == 0) {
            count++;
            break;
        }
    }
    if (count == 0)
        printf("Prime Number");
    else
        printf("Not Prime Number");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, count = 0;
    cout << "Enter any positive number: ";
    cin >> num;
    if (num <= 1) {
        cout << "Not Prime Number";
        return 0;
    }
    for (int i = 2; i < num; i++) {
        if (num % i == 0) {
            count++;
            break;
        }
    }
    if (count == 0)
        cout << "Prime Number";
    else
        cout << "Not Prime Number";
    return 0;
}
```

২২. পরিপূর্ণ সংখ্যা যাচাই 22. CHECK PERFECT NUMBER

পরিপূর্ণ সংখ্যা: নিজ ব্যতীত সকল গুণনীয়কের যোগফল = নিজেই।
উদাহরণ: 6 → গুণনীয়ক: 1,2,3 → 1+2+3 = 6 ✓
28 → গুণনীয়ক: 1,2,4,7,14 → যোগফল = 28 ✓
12 → গুণনীয়ক যোগ = 16 ≠ 12 ✗

C

```c
#include <stdio.h>

int main() {
    int num, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    for (int i = 1; i < num; i++) {
        if (num % i == 0)
            sum = sum + i;
    }
    if (sum == num)
        printf("Perfect Number\n");
    else
        printf("Not Perfect Number\n");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    for (int i = 1; i < num; i++) {
        if (num % i == 0)
            sum = sum + i;
    }
    if (sum == num)
        cout << "Perfect Number" << endl;
    else
        cout << "Not Perfect Number" << endl;
    return 0;
}
```

২৩. সংখ্যার অঙ্কসমূহের যোগফল 23. SUM OF DIGITS IN AN INTEGER

```
অঙ্কসমূহের যোগফল: % 10 দিয়ে শেষ অঙ্ক নেওয়া, / 10 দিয়ে কমানো।
  উদাহরণ: 1234 → r=4, sum=4, temp=123
               → r=3, sum=7, temp=12
               → r=2, sum=9, temp=1  → r=1, sum=10 ✓
```

C

```c
#include <stdio.h>

int main() {
    int num, r, sum = 0, temp;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum + r;
        temp = temp / 10;
    }
    printf("Sum of digits = %d", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, r, sum = 0, temp;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum + r;
        temp = temp / 10;
    }
    cout << "Sum of digits = " << sum;
    return 0;
}
```

২৪. সংখ্যা উল্টানো 24. REVERSE A NUMBER

```
সংখ্যা উল্টানো: rev = rev × 10 + (শেষ অঙ্ক)
  উদাহরণ: 1234 উল্টালে 4321
         rev=0 → r=4, rev=4 → r=3, rev=43 → r=2, rev=432 → r=1, rev=4321
```

C

```c
#include <stdio.h>

int main() {
    int num, r, sum = 0, temp;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum * 10 + r;
        temp = temp / 10;
    }
    printf("Reversed number = %d", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, r, sum = 0, temp;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum * 10 + r;
        temp = temp / 10;
    }
    cout << "Reversed number = " << sum;
    return 0;
}
```

২৫. প্যালিন্ড্রোম সংখ্যা যাচাই 25. CHECK PALINDROME NUMBER

```
প্যালিন্ড্রোম: সংখ্যা উল্টালেও একই হলে প্যালিন্ড্রোম।
  উদাহরণ: 121 উল্টালে 121 → প্যালিন্ড্রোম ✓
         12321 উল্টালে 12321 ✓  |  1234 উল্টালে 4321 ✗
```

C

```c
#include <stdio.h>

int main() {
    int num, temp, r, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum * 10 + r;
        temp = temp / 10;
    }
    if (num == sum)
        printf("Palindrome");
    else
        printf("Not a palindrome");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, temp, r, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        sum = sum * 10 + r;
        temp = temp / 10;
    }
    if (num == sum)
        cout << "Palindrome";
    else
        cout << "Not a palindrome";
    return 0;
}
```

২৬. আর্মস্ট্রং সংখ্যা যাচাই 26. CHECK ARMSTRONG NUMBER

আর্মস্ট্রং সংখ্যা: n-অঙ্কের সংখ্যায় প্রতিটি অঙ্কের n-তম ঘাতের যোগফল = সংখ্যা।
উদাহরণ: 153 (৩ অঙ্ক) → $1^3+5^3+3^3$ = 1+125+27 = 153 ✓
370 → $3^3+7^3+0^3$ = 27+343+0 = 370 ✓
123 → 1+8+27 = 36 ≠ 123 ✗

C

```c
#include <stdio.h>

int main() {
    int num, r, temp, sum = 0, digits = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    temp = num;
    while (temp != 0) {
        digits++;
        temp = temp / 10;
    }
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        int p = 1;
        for (int i = 0; i < digits; i++)
            p = p * r;
        sum = sum + p;
        temp = temp / 10;
    }
    if (sum == num)
        printf("Armstrong number");
    else
        printf("Not an Armstrong number");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, r, temp, sum = 0, digits = 0;
    cout << "Enter a number: ";
    cin >> num;
    temp = num;
    while (temp != 0) {
        digits++;
        temp = temp / 10;
    }
    temp = num;
    while (temp != 0) {
        r = temp % 10;
        int p = 1;
        for (int i = 0; i < digits; i++)
            p = p * r;
        sum = sum + p;
        temp = temp / 10;
    }
    if (sum == num)
        cout << "Armstrong number";
    else
        cout << "Not an Armstrong number";
    return 0;
}
```

২৭. স্ট্রং সংখ্যা যাচাই 27. CHECK STRONG NUMBER

স্ট্রং সংখ্যা: প্রতিটি অঙ্কের ফ্যাক্টরিয়ালের যোগফল = সংখ্যাটি।
উদাহরণ: 145 → 1!+4!+5! = 1+24+120 = 145 ✓
2 → 2! = 2 ✓ | 40585 → 4!+0!+5!+8!+5! = 40585 ✓

C

```c
#include <stdio.h>

int main() {
    int num, originalNum, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    originalNum = num;
    while (num > 0) {
        int digit = num % 10;
        int fact = 1;
        for (int i = 1; i <= digit; i++)
            fact = fact * i;
        sum = sum + fact;
        num = num / 10;
    }
    if (sum == originalNum)
        printf("Strong Number\n");
    else
        printf("Not Strong Number\n");
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num, originalNum, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    originalNum = num;
    while (num > 0) {
        int digit = num % 10;
        int fact = 1;
        for (int i = 1; i <= digit; i++)
            fact = fact * i;
        sum = sum + fact;
        num = num / 10;
    }
    if (sum == originalNum)
        cout << "Strong Number" << endl;
    else
        cout << "Not Strong Number" << endl;
    return 0;
}
```

গ.সা.গু, ল.সা.গু ও ফ্যাক্টরিয়াল — GCD, LCM & Factorial

২৮. গ.সা.গু এবং ল.সা.গু 28. GCD AND LCM OF TWO NUMBERS

```
গ.সা.গু (GCD): ইউক্লিড অ্যালগরিদম — n1 % n2 করতে থাকো, n2=0 হলে n1-ই GCD।
  ল.সা.গু (LCM) = (a / GCD) × b   ← আগে ভাগ, তারপর গুণ (overflow এড়াতে)
  উদাহরণ: a=12, b=18 → GCD=6 → LCM = (12/6)×18 = 2×18 = 36
```

C

```c
#include <stdio.h>

int main() {
    int num1, num2, n1, n2, rem, gcd, lcm;
    printf("Enter two numbers: ");
    scanf("%d %d", &num1, &num2);
    n1 = num1;
    n2 = num2;
    while (n2 != 0) {
        rem = n1 % n2;
        n1 = n2;
        n2 = rem;
    }
    gcd = n1;
    lcm = (num1 / gcd) * num2;
    printf("GCD = %d\nLCM = %d", gcd, lcm);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num1, num2, n1, n2, rem, gcd, lcm;
    cout << "Enter two numbers: ";
    cin >> num1 >> num2;
    n1 = num1;
    n2 = num2;
    while (n2 != 0) {
        rem = n1 % n2;
        n1 = n2;
        n2 = rem;
    }
    gcd = n1;
    lcm = (num1 / gcd) * num2;
    cout << "GCD = " << gcd << "\nLCM = " << lcm;
    return 0;
}
```

২৯. ফ্যাক্টরিয়াল নির্ণয় 29. FACTORIAL OF A NUMBER

```
ফ্যাক্টরিয়াল: n! = 1 × 2 × 3 × ... × n
  উদাহরণ: 5! = 1×2×3×4×5 = 120
          0! = 1 (সংজ্ঞা অনুসারে)
  লুপে: fact=1, i=1 থেকে n পর্যন্ত → fact = fact × i
```

C

```c
#include <stdio.h>

int main() {
    int num;
    unsigned long long fact = 1;
    printf("Enter a number: ");
    scanf("%d", &num);
    for (int i = 1; i <= num; i++)
        fact = fact * i;
    printf("Factorial = %llu", fact); // %llu → unsigned long l
ong (বড় সংখ্যা)
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    unsigned long long fact = 1;
    cout << "Enter a number: ";
    cin >> num;
    for (int i = 1; i <= num; i++)
        fact = fact * i;
    cout << "Factorial = " << fact;
    return 0;
}
```

সিরিজ ও যোগফল — Series & Summation

৩০. ফিবোনাচ্চি সিরিজ 30. FIBONACCI SERIES

```
ফিবোনাচ্চি সিরিজ: প্রতিটি সংখ্যা আগের দুটির যোগফল।
  সিরিজ: 0  1  1  2  3  5  8  13  21  34 ...
  নিয়ম: a=0, b=1 → পরের সংখ্যা c=a+b, তারপর a=b, b=c
```

C

```c
#include <stdio.h>

int main() {
    int n, first = 0, second = 1, next;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    if (n >= 1) printf("%d ", first);
    if (n >= 2) printf("%d ", second);
    for (int i = 3; i <= n; i++) {
        next = first + second;
        printf("%d ", next);
        first = second;
        second = next;
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, first = 0, second = 1, next;
    cout << "Enter number of terms: ";
    cin >> n;
    if (n >= 1) cout << first << " ";
    if (n >= 2) cout << second << " ";
    for (int i = 3; i <= n; i++) {
        next = first + second;
        cout << next << " ";
        first = second;
        second = next;
    }
    return 0;
}
```

৩১. সিরিজ: ১! + ২! + ৩! + ... + n! 31. SERIES: 1! + 2! + 3! + 4! + ... + n! = ?

C

```c
#include <stdio.h>

int main() {
    int num;
    unsigned long long fact = 1, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    for (int i = 1; i <= num; i++) {
        fact = fact * i;
        sum = sum + fact;
    }
    printf("Sum = %llu", sum); // %llu → unsigned long long (বড় সংখ্যা)
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    unsigned long long fact = 1, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    for (int i = 1; i <= num; i++) {
        fact = fact * i;
        sum = sum + fact;
    }
    cout << "Sum = " << sum;
    return 0;
}
```

৩২. সিরিজ: ২! + ৪! + ৬! + ... + n! 32. SERIES: 2! + 4! + 6! + ... + n! = ?

C

```c
#include <stdio.h>

int main() {
    int num;
    unsigned long long fact = 1, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    for (int i = 2; i <= num; i = i + 2) {
        fact = 1;
        for (int j = 1; j <= i; j++)
            fact = fact * j;
        sum = sum + fact;
    }
    printf("Sum = %llu", sum); // %llu → unsigned long long (বড় সংখ্যা)
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    unsigned long long fact = 1, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    for (int i = 2; i <= num; i = i + 2) {
        fact = 1;
        for (int j = 1; j <= i; j++)
            fact = fact * j;
        sum = sum + fact;
    }
    cout << "Sum = " << sum;
    return 0;
}
```

৩৩. সিরিজ: ১! + ৩! + ৫! + ... + n! 33. SERIES: 1! + 3! + 5! + ... + n! = ?

C

```c
#include <stdio.h>

int main() {
    int num;
    unsigned long long fact = 1, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    for (int i = 1; i <= num; i = i + 2) {
        fact = 1;
        for (int j = 1; j <= i; j++)
            fact = fact * j;
        sum = sum + fact;
    }
    printf("Sum = %llu", sum); // %llu → unsigned long long (বড় সংখ্যা)
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    unsigned long long fact = 1, sum = 0;
    cout << "Enter a number: ";
    cin >> num;
    for (int i = 1; i <= num; i = i + 2) {
        fact = 1;
        for (int j = 1; j <= i; j++)
            fact = fact * j;
        sum = sum + fact;
    }
    cout << "Sum = " << sum;
    return 0;
}
```

৩৪. সিরিজ: ১ + ২ + ৩ + ... + n 34. SERIES: 1 + 2 + 3 + 4 + ... + n = ?

C

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++)
        sum = sum + i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        sum = sum + i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৫. সিরিজ: ১ + ৩ + ৫ + ... + n 35. SERIES: 1 + 3 + 5 + ... + n = ?

C

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i = i + 2)
        sum = sum + i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i = i + 2)
        sum = sum + i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৬. সিরিজ: ২ + ৪ + ৬ + ৮ + ... + n 36. SERIES: 2 + 4 + 6 + 8 + ... + n = ?

C

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 2; i <= n; i = i + 2)
        sum = sum + i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 2; i <= n; i = i + 2)
        sum = sum + i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৭. সিরিজ: ১^1 + ২^2 + ৩^3 + ... + n^n 37. SERIES: $1^1 + 2^2 + 3^3 + 4 + ... + n^n$ = ?

C

```c
#include <stdio.h>
#include <math.h>

int main() {
    int n;
    long long sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++)
        sum = sum + (long long)pow(i, i);
    printf("Sum = %lld\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main() {
    int n;
    long long sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        sum = sum + (long long)pow(i, i);
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৮. সিরিজ: ১^2 + ২^2 + ৩^2 + ... + n^2 38. SERIES: $1^2 + 2^2 + 3^2 + 4^2 + ... + n^2$ = ?

C

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++)
        sum = sum + i * i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        sum = sum + i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৩৯. সিরিজ: ১² + ৩² + ৫² + ... + n² 39. SERIES: $1^2 + 3^2 + 5^2 + \ldots + n^2 = ?$

C

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i = i + 2)
        sum = sum + i * i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i = i + 2)
        sum = sum + i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৪০. সিরিজ: ২² + ৪² + ৬² + ৮² + ... + n² 40. SERIES: $2^2 + 4^2 + 6^2 + 8^2 + \ldots + n^2 = ?$

C

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 2; i <= n; i = i + 2)
        sum = sum + i * i;
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 2; i <= n; i = i + 2)
        sum = sum + i * i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৪১. সিরিজ: ১ + ২ + ৪ + ৮ + ... + 2ⁿ 41. SERIES: $1 + 2 + 4 + 8 + \ldots + 2^n = ?$

C

```c
#include <stdio.h>

int main() {
    int n, sum = 0, term = 1;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        sum = sum + term;
        term = term * 2;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0, term = 1;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 0; i < n; i++) {
        sum = sum + term;
        term = term * 2;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৪২. সিরিজ: ১ + ৩ + ৯ + ২৭ + ... + 3ⁿ 42. SERIES: $1 + 3 + 9 + 27 + \ldots + 3^n = ?$

C

```c
#include <stdio.h>

int main() {
    int n, sum = 0, term = 1;
    printf("Enter number of terms: ");
    scanf("%d", &n);
    for (int i = 0; i < n; i++) {
        sum = sum + term;
        term = term * 3;
    }
    printf("Sum = %d\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0, term = 1;
    cout << "Enter number of terms: ";
    cin >> n;
    for (int i = 0; i < n; i++) {
        sum = sum + term;
        term = term * 3;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৪৩. সিরিজ: ১/১ + ১/২ + ১/৩ + ... + ১/n 43. SERIES: 1/1 + 1/2 + 1/3 + ... + 1/n = ?

C

```c
#include <stdio.h>

int main() {
    int n;
    double sum = 0.0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++)
        sum = sum + 1.0 / i;
    printf("Sum = %lf\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    double sum = 0.0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        sum = sum + 1.0 / i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

৪৪. সিরিজ: $১/১^2 + ১/২^2 + ১/৩^2 + ... + ১/n^2$ 44. SERIES: $1/1^2 + 1/2^2 + 1/3^2 + ... + 1/n^2$ = ?

C

```c
#include <stdio.h>

int main() {
    int n;
    double sum = 0.0;
    printf("Enter n: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++)
        sum = sum + 1.0 / (i * i);
    printf("Sum = %lf\n", sum);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    double sum = 0.0;
    cout << "Enter n: ";
    cin >> n;
    for (int i = 1; i <= n; i++)
        sum = sum + 1.0 / (i * i);
    cout << "Sum = " << sum << endl;
    return 0;
}
```

প্যাটার্ন প্রোগ্রাম — Pattern Programs

সব প্যাটার্নে একই নেস্টেড লুপ কাঠামো — শুধু **আউটপুট এক্সপ্রেশন** পরিবর্তন হয়।

৪৫. গ্রুপ A — বাম থেকে ক্রমবর্ধমান ত্রিভুজ (n = 5) — টেমপ্লেট (টাইপ ১-৮)

45. GROUP A — ASCENDING LEFT TRIANGLE — GENERAL TEMPLATE (TYPES 1-8)

```
printf("* ")
cout << "* "

*
* *
* * *
* * * *
* * * * *
```

```
printf("# ")
cout << "# "

#
# #
# # #
# # # #
# # # # #
```

```
printf("%d ", col)
cout << col << " "

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
printf("%d ", row)
cout << row << " "

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

```
printf("%d ", col%2)
cout << col%2 << " "

1
1 0
1 0 1
1 0 1 0
1 0 1 0 1
```

```
printf("%d ", row%2)
cout << row%2 << " "

1
0 0
1 1 1
0 0 0 0
1 1 1 1 1
```

```
printf("%c ", row+64)
cout << (char)(row+64) << " "

A
B B
C C C
D D D D
E E E E E
```

```
printf("%c ", col+64)
cout << (char)(col+64) << " "

A
A B
A B C
A B C D
A B C D E
```

C

```c
#include <stdio.h>
int main() {
    int row, col, n;
    printf("Enter a value of n: ");
    scanf("%d", &n);
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= row; col++) {
            printf(" "); // ← এখানে expression পরিবর্তন করো (উপ
রের টেবিল দেখো)
        }
        printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    int row, col, n;
    cout << "Enter a value of n: ";
    cin >> n;
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= row; col++)  {
            cout << " "; // ← এখানে expression পরিবর্তন করো (উপ
রের টেবিল দেখো)
        }
        cout << endl;
    }
    return 0;
}
```

৪৬. গ্রুপ B — বাম থেকে ক্রমহ্রাসমান ত্রিভুজ — টেমপ্লেট (টাইপ ৯–১৬) 46. GROUP B — DESCENDING LEFT TRIANGLE TEMPLATE (TYPES 9–16)

```
printf("* ")
cout << "* "
```

```
* * * * *
* * * *
* * *
* *
*
```

```
printf("# ")
cout << "# "
```

```
# # # # #
# # # #
# # #
# #
#
```

```
printf("%d ", col)
cout << col << " "
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
printf("%d ", row)
cout << row << " "
```

```
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

```
printf("%d ", col%2)
cout << col%2 << " "
```

```
1 0 1 0 1
1 0 1 0
1 0 1
1 0
1
```

```
printf("%d ", row%2)
cout << row%2 << " "
```

```
1 1 1 1 1
0 0 0 0
1 1 1
0 0
1
```

```
printf("%c ", row+64)
cout << (char)(row+64) << " "
```

```
E E E E E
D D D D
C C C
B B
A
```

```
printf("%c ", col+64)
cout << (char)(col+64) << " "
```

```
A B C D E
A B C D
A B C
A B
A
```

C

```c
#include <stdio.h>
int main() {
    int row, col, n;
    printf("Enter a value of n: ");
    scanf("%d", &n);
    for (row = n; row >= 1; row--) {
        for (col = 1; col <= row; col++) {
            printf(" "); // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        }
        printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    int row, col, n;
    cout << "Enter a value of n: ";
    cin >> n;
    for (row = n; row >= 1; row--) {
        for (col = 1; col <= row; col++) {
            cout << " "; // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        }
        cout << endl;
    }
    return 0;
}
```

৪৭. গ্রুপ C — ডায়মন্ড আকৃতি — টেমপ্লেট (টাইপ ১৭–২৪) 47. GROUP C — DIAMOND SHAPE TEMPLATE (TYPES 17–24)

```
printf("* ")
cout << "* "
```

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

```
printf("# ")
cout << "# "
```

```
#
# #
# # #
# # # #
# # # # #
# # # #
# # #
# #
#
```

```
printf("%d ", col)
cout << col << " "
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
printf("%d ", row)
cout << row << " "
```

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

```
printf("%d ", col%2)
cout << col%2 << " "
```

```
1
1 0
1 0 1
1 0 1 0
1 0 1 0 1
1 0 1 0
1 0 1
1 0
1
```

```
printf("%d ", row%2)
cout << row%2 << " "
```

```
1
0 0
1 1 1
0 0 0 0
1 1 1 1 1
0 0 0 0
1 1 1
0 0
1
```

```
printf("%c ", row+64)
cout << (char)(row+64) << " "
```

```
A
B B
C C C
D D D D
E E E E E
D D D D
C C C
B B
A
```

```
printf("%c ", col+64)
cout << (char)(col+64) << " "
```

```
A
A B
A B C
A B C D
A B C D E
A B C D
A B C
A B
A
```

C

```c
#include <stdio.h>
int main() {
    int row, col, n;
    printf("Enter a value of n: ");
    scanf("%d", &n);
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= row; col++)
            printf(" "); // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        printf("\n");
    }
    for (row = n - 1; row >= 1; row--) {
        for (col = 1; col <= row; col++)
            printf(" "); // ← একই expression এখানেও রাখো (নিচের অর্ধেক)
        printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    int row, col, n;
    cout << "Enter a value of n: ";
    cin >> n;
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= row; col++)
            cout << " "; // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        cout << endl;
    }
    for (row = n - 1; row >= 1; row--) {
        for (col = 1; col <= row; col++)
            cout << " "; // ← একই expression এখানেও রাখো (নিচের অর্ধেক)
        cout << endl;
    }
    return 0;
}
```

৪৮. গ্রুপ D — ডানদিকে সাজানো ক্রমবর্ধমান ত্রিভুজ — টেমপ্লেট (টাইপ ২৫–৩২)

48. GROUP D — RIGHT-ALIGNED ASCENDING TRIANGLE TEMPLATE (TYPES 25-32)

```
printf("* ")
cout << "* "

        *
      * *
    * * *
  * * * *
* * * * *
```

```
printf("# ")
cout << "# "

        #
      # #
    # # #
  # # # #
# # # # #
```

```
printf("%d ", col)
cout << col << " "

        1
      1 2
    1 2 3
  1 2 3 4
1 2 3 4 5
```

```
printf("%d ", row)
cout << row << " "

        1
      2 2
    3 3 3
  4 4 4 4
5 5 5 5 5
```

```
printf("%d ", col%2)
cout << col%2 << " "

        1
      1 0
    1 0 1
  1 0 1 0
1 0 1 0 1
```

```
printf("%d ", row%2)
cout << row%2 << " "

        1
      0 0
    1 1 1
  0 0 0 0
1 1 1 1 1
```

```
printf("%c ", row+64)
cout << (char)(row+64) << " "

        A
      B B
    C C C
  D D D D
E E E E E
```

```
printf("%c ", col+64)
cout << (char)(col+64) << " "

        A
      A B
    A B C
  A B C D
A B C D E
```

C

```c
#include <stdio.h>
int main() {
    int row, col, n;
    printf("Enter a value of n: ");
    scanf("%d", &n);
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= n - row; col++)
            printf("  "); // ← ফাঁকা জায়গা তৈরি করে ডানে সরাতে (পরিবর্তন করো না)
        for (col = 1; col <= row; col++)
            printf(" "); // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    int row, col, n;
    cout << "Enter a value of n: ";
    cin >> n;
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= n - row; col++)
            cout << "  "; // ← ফাঁকা জায়গা তৈরি করে ডানে সরাতে (পরিবর্তন করো না)
        for (col = 1; col <= row; col++)
            cout << " "; // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        cout << endl;
    }
    return 0;
}
```

৪৯. গ্রুপ E — ডানদিকে সাজানো ক্রমহ্রাসমান ত্রিভুজ — টেমপ্লেট (টাইপ ৩৩–৪০)

49. GROUP E — RIGHT-ALIGNED DESCENDING TRIANGLE TEMPLATE (TYPES 33-40)

```
printf("* ")
cout << "* "

* * * * *
  * * * *
    * * *
      * *
        *
```

```
printf("# ")
cout << "# "

# # # # #
  # # # #
    # # #
      # #
        #
```

```
printf("%d ", col)
cout << col << " "

1 2 3 4 5
  1 2 3 4
    1 2 3
      1 2
        1
```

```
printf("%d ", row)
cout << row << " "

5 5 5 5 5
  4 4 4 4
    3 3 3
      2 2
        1
```

```
printf("%d ", col%2)
cout << col%2 << " "

1 0 1 0 1
  1 0 1 0
    1 0 1
      1 0
        1
```

```
printf("%d ", row%2)
cout << row%2 << " "

1 1 1 1 1
  0 0 0 0
    1 1 1
      0 0
        1
```

```
printf("%c ", row+64)
cout << (char)(row+64) << " "

E E E E E
  D D D D
    C C C
      B B
        A
```

```
printf("%c ", col+64)
cout << (char)(col+64) << " "

A B C D E
  A B C D
    A B C
      A B
        A
```

C

```c
#include <stdio.h>
int main() {
    int row, col, n;
    printf("Enter a value of n: ");
    scanf("%d", &n);
    for (row = n; row >= 1; row--) {
        for (col = 1; col <= n - row; col++)
            printf("  "); // ← ফাঁকা জায়গা তৈরি করে ডানে সরাতে (পরিবর্তন করো না)
        for (col = 1; col <= row; col++)
            printf(" "); // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    int row, col, n;
    cout << "Enter a value of n: ";
    cin >> n;
    for (row = n; row >= 1; row--) {
        for (col = 1; col <= n - row; col++)
            cout << "  "; // ← ফাঁকা জায়গা তৈরি করে ডানে সরাতে (পরিবর্তন করো না)
        for (col = 1; col <= row; col++)
            cout << " "; // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        cout << endl;
    }
    return 0;
}
```

৫০. গ্রুপ F — আয়তক্ষেত্র / বর্গক্ষেত্র (n×n) — টেমপ্লেট (টাইপ ৪১–৪৮) 50. GROUP F — RECTANGLE / SQUARE (n×n) TEMPLATE (TYPES 41–48)

```
printf("* ")
cout << "* "
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
printf("# ")
cout << "# "
```

```
# # # # #
# # # # #
# # # # #
# # # # #
# # # # #
```

```
printf("%d ", col)
cout << col << " "
```

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

```
printf("%d ", row)
cout << row << " "
```

```
1 1 1 1 1
2 2 2 2 2
3 3 3 3 3
4 4 4 4 4
5 5 5 5 5
```

```
printf("%d ", col%2)
cout << col%2 << " "
```

```
1 0 1 0 1
1 0 1 0 1
1 0 1 0 1
1 0 1 0 1
1 0 1 0 1
```

```
printf("%d ", row%2)
cout << row%2 << " "
```

```
1 1 1 1 1
0 0 0 0 0
1 1 1 1 1
0 0 0 0 0
1 1 1 1 1
```

```
printf("%c ", row+64)
cout << (char)(row+64) << " "
```

```
A A A A A
B B B B B
C C C C C
D D D D D
E E E E E
```

```
printf("%c ", col+64)
cout << (char)(col+64) << " "
```

```
A B C D E
A B C D E
A B C D E
A B C D E
A B C D E
```

C

```c
#include <stdio.h>
int main() {
    int row, col, n;
    printf("Enter a value of n: ");
    scanf("%d", &n);
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= n; col++)
            printf(" "); // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    int row, col, n;
    cout << "Enter a value of n: ";
    cin >> n;
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= n; col++)
            cout << " "; // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        cout << endl;
    }
    return 0;
}
```

৫১. গ্রুপ G — গণনা প্যাটার্ন — টেমপ্লেট (টাইপ ৪৯–৫১) 51. GROUP G — COUNTING PATTERNS TEMPLATE (TYPES 49–51)

```
printf("%d ", ++count)
cout << ++count << " "
```

```
1
2 3
4 5 6
7 8 9 10
```

```
printf("%d ", count++)
cout << count++ << " "
```

```
0
1 2
3 4 5
6 7 8 9
```

```
printf("%d ", col*row)
cout << col*row << " "
```

```
1
2 4
3 6 9
4 8 12 16
```

C

```c
#include <stdio.h>
int main() {
    int row, col, count = 0, n;
    printf("Enter a value of n: ");
    scanf("%d", &n);
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= row; col++) {
            printf(" "); // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        }
        printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    int row, col, count = 0, n;
    cout << "Enter a value of n: ";
    cin >> n;
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= row; col++) {
            cout << " "; // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        }
        cout << endl;
    }
    return 0;
}
```

৫২. গ্রুপ H — কেন্দ্রীভূত পিরামিড — টেমপ্লেট (টাইপ ৫২–৫৯) 52. GROUP H — CENTERED PYRAMID TEMPLATE (TYPES 52-59)

```
printf("* ")
cout << "* "

    *
   * * *
  * * * * *
 * * * * * * *
* * * * * * * * *
```

```
printf("# ")
cout << "# "

    #
   # # #
  # # # # #
 # # # # # # #
# # # # # # # # #
```

```
printf("%d ", col)
cout << col << " "

    1
   1 2 3
  1 2 3 4 5
 1 2 3 4 5 6 7
1 2 3 4 5 6 7 8 9
```

```
printf("%d ", row)
cout << row << " "

    1
   2 2 2
  3 3 3 3 3
 4 4 4 4 4 4 4
5 5 5 5 5 5 5 5 5
```

```
printf("%d ", col%2)
cout << col%2 << " "

    1
   1 0 1
  1 0 1 0 1
 1 0 1 0 1 0 1
1 0 1 0 1 0 1 0 1
```

```
printf("%d ", row%2)
cout << row%2 << " "

    1
   0 0 0
  1 1 1 1 1
 0 0 0 0 0 0 0
1 1 1 1 1 1 1 1 1
```

```
printf("%c ", row+64)
cout << (char)(row+64) << " "

    A
   B B B
  C C C C C
 D D D D D D D
E E E E E E E E E
```

```
printf("%c ", col+64)
cout << (char)(col+64) << " "

    A
   A B C
  A B C D E
 A B C D E F G
A B C D E F G H I
```

C

```c
#include <stdio.h>
int main() {
    int row, col, n;
    printf("Enter a value of n: ");
    scanf("%d", &n);
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= n - row; col++)
            printf(" ");              // ← পিরামিডকে মাঝখানে আনতে ফাঁকা (পরিবর্তন করো না)
        for (col = 1; col <= 2 * row - 1; col++)
            printf(" "); // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        printf("\n");
    }
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;
int main() {
    int row, col, n;
    cout << "Enter a value of n: ";
    cin >> n;
    for (row = 1; row <= n; row++) {
        for (col = 1; col <= n - row; col++)
            cout << " ";              // ← পিরামিডকে মাঝখানে আনতে ফাঁকা (পরিবর্তন করো না)
        for (col = 1; col <= 2 * row - 1; col++)
            cout << " "; // ← এখানে expression পরিবর্তন করো (উপরের টেবিল দেখো)
        cout << endl;
    }
    return 0;
}
```

রিকার্শন — Recursion

৫৩. রিকার্শন দিয়ে ফ্যাক্টরিয়াল 53. FACTORIAL USING RECURSION

```
রিকার্শন দিয়ে ফ্যাক্টরিয়াল:
  fact(5) = 5 × fact(4) = 5 × 4 × fact(3)
          = 5 × 4 × 3 × fact(2) = 5 × 4 × 3 × 2 × fact(1)
          = 5 × 4 × 3 × 2 × 1 = 120
  Base case: fact(1) = 1  ← এখানে রিকার্শন থামে।
```

C

```c
#include <stdio.h>

int fact(int n) {
    if (n <= 1) return 1;
    else return n * fact(n - 1);
}

int main() {
    int number, result;
    printf("Enter a positive number: ");
    scanf("%d", &number);
    result = fact(number);
    printf("%d", result);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int fact(int n) {
    if (n <= 1) return 1;
    else return n * fact(n - 1);
}

int main() {
    int number, result;
    cout << "Enter a positive number: ";
    cin >> number;
    result = fact(number);
    cout << result;
    return 0;
}
```

৫৪. রিকার্শন দিয়ে ফিবোনাচ্চি সিরিজ 54. FIBONACCI SERIES USING RECURSION

```
রিকার্শন দিয়ে ফিবোনাচ্চি: fibo(n) = fibo(n-1) + fibo(n-2)
  Base cases: fibo(0)=0, fibo(1)=1
  fibo(5) = fibo(4)+fibo(3) = (fibo(3)+fibo(2))+(fibo(2)+fibo(1))
          = ... = 3+2 = 5
  সিরিজ: 0  1  1  2  3  5  8  13 ...
```

C

```c
#include <stdio.h>

int fibo(int n) {
    if (n == 0) return 0;
    else if (n == 1) return 1;
    else return fibo(n-1) + fibo(n-2);
}

int main() {
    int terms;
    printf("Enter the number of terms: ");
    scanf("%d", &terms);
    for (int i = 0; i < terms; i++)
        printf("%d ", fibo(i));
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int fibo(int n) {
    if (n == 0) return 0;
    else if (n == 1) return 1;
    else return fibo(n-1) + fibo(n-2);
}

int main() {
    int terms;
    cout << "Enter the number of terms: ";
    cin >> terms;
    for (int i = 0; i < terms; i++)
        cout << fibo(i) << " ";
    return 0;
}
```

৫৫. রিকার্শন দিয়ে ঘাত (পাওয়ার) নির্ণয় 55. CALCULATE POWER USING RECURSION

```
রিকার্শনে পাওয়ার: power(x,y) = x × power(x, y-1)
  Base case: power(x, 0) = 1
  উদাহরণ: power(2,4) = 2×power(2,3) = 2×2×power(2,2)
          = 2×2×2×power(2,1) = 2×2×2×2×power(2,0) = 2×2×2×2×1 = 16
```

C

```c
#include <stdio.h>

int power(int x, int y) {
    if (y == 0) return 1;
    else return (x * power(x, y-1));
}

int main() {
    int b, p, c;
    printf("Enter Base and Power: ");
    scanf("%d %d", &b, &p);
    c = power(b, p);
    printf("%d", c);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int power(int x, int y) {
    if (y == 0) return 1;
    else return (x * power(x, y-1));
}

int main() {
    int b, p, c;
    cout << "Enter Base and Power: ";
    cin >> b >> p;
    c = power(b, p);
    cout << c;
    return 0;
}
```

অ্যারে — Arrays

৫৬. অ্যারেতে সর্বোচ্চ মান খোঁজা 56. FIND MAXIMUM IN AN ARRAY

C

```c
#include <stdio.h>

int main() {
    int num[100], n, i;
    printf("How many numbers: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) scanf("%d", &num[i]);
    int max = num[0];
    for (i = 0; i < n; i++)
        if (max < num[i]) max = num[i];
    printf("%d\n", max);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num[100], n, i;
    cout << "How many numbers: ";
    cin >> n;
    for (i = 0; i < n; i++) cin >> num[i];
    int max = num[0];
    for (i = 0; i < n; i++)
        if (max < num[i]) max = num[i];
    cout << max << endl;
    return 0;
}
```

৫৭. অ্যারেতে সর্বনিম্ন মান খোঁজা 57. FIND MINIMUM IN AN ARRAY

C

```c
#include <stdio.h>

int main() {
    int num[100], n, i;
    printf("How many numbers: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) scanf("%d", &num[i]);
    int min = num[0];
    for (i = 0; i < n; i++)
        if (min > num[i]) min = num[i];
    printf("%d\n", min);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int num[100], n, i;
    cout << "How many numbers: ";
    cin >> n;
    for (i = 0; i < n; i++) cin >> num[i];
    int min = num[0];
    for (i = 0; i < n; i++)
        if (min > num[i]) min = num[i];
    cout << min << endl;
    return 0;
}
```

বিবিধ — Miscellaneous

৫৮. টেম্প ভেরিয়েবল ছাড়া দুটি সংখ্যা অদলবদল 58. SWAP TWO NUMBERS WITHOUT TEMPORARY VARIABLE

```
তৃতীয় চলক (temp) ছাড়া অদলবদল — গাণিতিক কৌশল:
  a = 5,  b = 3
  a = a + b = 8      (a-তে যোগফল রাখা হলো)
  b = a - b = 8-3=5  (পুরনো a পাওয়া গেল)
  a = a - b = 8-5=3  (পুরনো b পাওয়া গেল)
  ফলাফল: a = 3,  b = 5  ✓
```

C

```c
#include <stdio.h>

int main() {
    int a, b;
    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);
    a = b + a;
    b = a - b;
    a = a - b;
    printf("%d %d", a, b);
    return 0;
}
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    a = b + a;
    b = a - b;
    a = a - b;
    cout << a << " " << b;
    return 0;
}
```

লুপ নিয়ন্ত্রণ — Loop Control

৫৯. continue স্টেটমেন্ট — ৩ বাদ দিয়ে বাকি প্রিন্ট 59. CONTINUE STATEMENT — SKIP 3, PRINT REST

```
continue কী করে?
লুপের ভেতরে continue পেলে সেই পুনরাবৃত্তির বাকি অংশ বাদ দিয়ে
সরাসরি পরের পুনরাবৃত্তিতে চলে যায়।
উদাহরণ (i==3 হলে skip): আউটপুট → 1 2 4 5 6 7 8 9 10  (3 বাদ)
```

C

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 3) continue;
        printf("%d ", i);
    }
    return 0;
} // Output: 1 2 4 5 6 7 8 9 10
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 3) continue;
        cout << i << " ";
    }
    return 0;
} // Output: 1 2 4 5 6 7 8 9 10
```

৬০. break স্টেটমেন্ট — ৩-এ থামানো 60. BREAK STATEMENT — STOP AT 3

```
break কী করে?
লুপের ভেতরে break পেলে সম্পূর্ণ লুপ বন্ধ হয়ে বাইরে চলে আসে।
উদাহরণ (i==3 হলে stop): আউটপুট → 1 2  (3 আসতেই থামে)

continue vs break:
  continue → শুধু এই পুনরাবৃত্তি বাদ, লুপ চলতে থাকে।
  break    → পুরো লুপই বন্ধ।
```

C

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 3) break;
        printf("%d ", i);
    }
    return 0;
} // Output: 1 2
```

C++

```cpp
#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 3) break;
        cout << i << " ";
    }
    return 0;
} // Output: 1 2
```